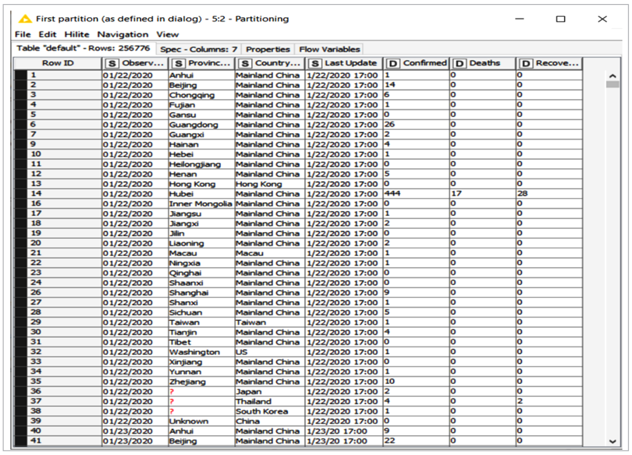
Partitioning node

Using the partitioning node, the given input table is divided into two different partitions row-wise. The first partition is termed *train data* and the second is *test data*. The outcomes of these two partitions are made available at the two output ports of the partitioning node. The data set has been partitioned in a 70:30 ratio, with 70 per cent of the data forming the training set and the remaining 30 per cent forming the testing data set.

The output of the first partition of the Covid 19 data set, also referred to as the training data set, is captured in Figure 4.



Row ID	S Observ...	S Provinc...	S Country...	S Last Update	D Confirmed	D Deaths	D Recove...
1	01/22/2020	Anhui	Mainland China	1/22/2020 17:00	1	0	0
2	01/22/2020	Beijing	Mainland China	1/22/2020 17:00	14	0	0
3	01/22/2020	Chongqing	Mainland China	1/22/2020 17:00	6	0	0
4	01/22/2020	Fujian	Mainland China	1/22/2020 17:00	1	0	0
5	01/22/2020	Gansu	Mainland China	1/22/2020 17:00	0	0	0
6	01/22/2020	Guangdong	Mainland China	1/22/2020 17:00	26	0	0
7	01/22/2020	Guangxi	Mainland China	1/22/2020 17:00	2	0	0
9	01/22/2020	Hainan	Mainland China	1/22/2020 17:00	4	0	0
10	01/22/2020	Hebei	Mainland China	1/22/2020 17:00	1	0	0
11	01/22/2020	Heilongjiang	Mainland China	1/22/2020 17:00	0	0	0
12	01/22/2020	Henan	Mainland China	1/22/2020 17:00	5	0	0
13	01/22/2020	Hong Kong	Hong Kong	1/22/2020 17:00	0	0	0
14	01/22/2020	Hubei	Mainland China	1/22/2020 17:00	444	17	28
16	01/22/2020	Inner Mongolia	Mainland China	1/22/2020 17:00	0	0	0
17	01/22/2020	Jiangsu	Mainland China	1/22/2020 17:00	1	0	0
18	01/22/2020	Jiangxi	Mainland China	1/22/2020 17:00	2	0	0
19	01/22/2020	Jilin	Mainland China	1/22/2020 17:00	0	0	0
20	01/22/2020	Liaoning	Mainland China	1/22/2020 17:00	2	0	0
21	01/22/2020	Macau	Macau	1/22/2020 17:00	1	0	0
22	01/22/2020	Ningxia	Mainland China	1/22/2020 17:00	1	0	0
23	01/22/2020	Qinghai	Mainland China	1/22/2020 17:00	0	0	0
24	01/22/2020	Shaanxi	Mainland China	1/22/2020 17:00	0	0	0
26	01/22/2020	Shanghai	Mainland China	1/22/2020 17:00	9	0	0
27	01/22/2020	Shanxi	Mainland China	1/22/2020 17:00	1	0	0
28	01/22/2020	Sichuan	Mainland China	1/22/2020 17:00	5	0	0
29	01/22/2020	Taiwan	Taiwan	1/22/2020 17:00	1	0	0
30	01/22/2020	Tianjin	Mainland China	1/22/2020 17:00	4	0	0
31	01/22/2020	Tibet	Mainland China	1/22/2020 17:00	0	0	0
32	01/22/2020	Washington	US	1/22/2020 17:00	1	0	0
33	01/22/2020	Xinjiang	Mainland China	1/22/2020 17:00	0	0	0
34	01/22/2020	Yunnan	Mainland China	1/22/2020 17:00	1	0	0
35	01/22/2020	Zhejiang	Mainland China	1/22/2020 17:00	10	0	0
36	01/22/2020	?	Japan	1/22/2020 17:00	2	0	0
37	01/22/2020	?	Thailand	1/22/2020 17:00	4	0	2
38	01/22/2020	?	South Korea	1/22/2020 17:00	1	0	0
39	01/22/2020	Unknown	China	1/22/2020 17:00	0	0	0
40	01/23/2020	Anhui	Mainland China	1/23/20 17:00	9	0	0
41	01/23/2020	Beijing	Mainland China	1/23/20 17:00	22	0	0

Figure 4: First partition of the training data set

K-means node

Based on the number of clusters (k) configured, the k-means node groups the input data set into the required cluster constructed on the distance criterion. This node outputs the cluster centres for a predefined number of clusters (no dynamic number of clusters).

K-means performs a crisp clustering that assigns a data vector to exactly one cluster. The algorithm terminates when the cluster assignments do not change any more. The clustering algorithm uses the Euclidean distance on the selected attributes. The data is not normalised by the node.

As the number of the cluster has been configured with the value '3', three clusters (cluster_0, cluster_1, cluster_2) have been created for the given Corona virus data set, as shown in Figure 6.

Clustering is based on three attributes — confirmed cases, cured (recovered) cases, and deaths. The number in the figure denotes cluster number labels like:

- CLUSTER 0 – Cases of COVID-19 deaths
- CLUSTER 1 – Cases of COVID-19 cured cases
- CLUSTER 2 – Cases of COVID-19 confirmed cases

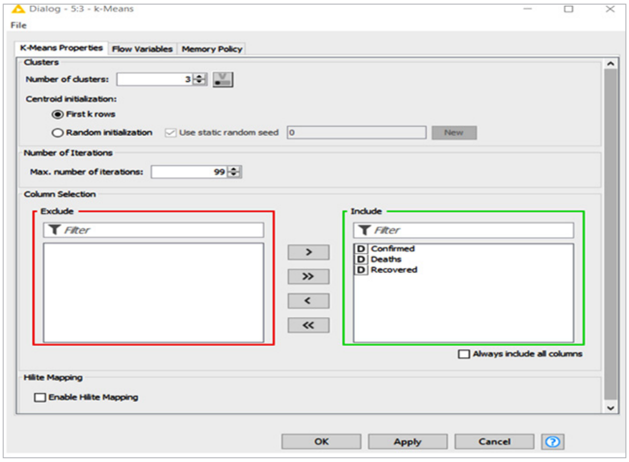


Figure 5: Configuration of k-means node

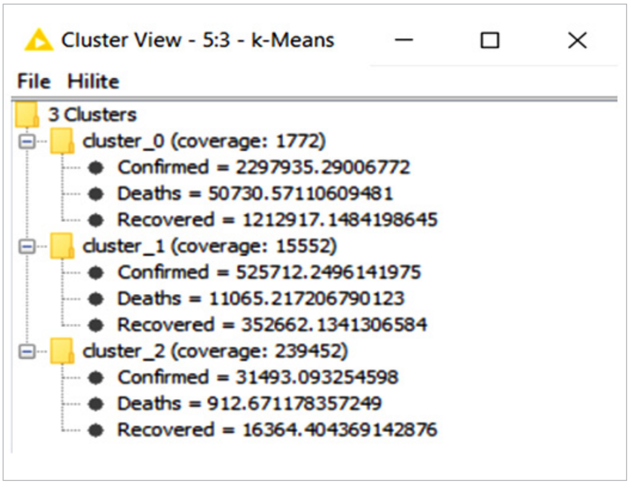


Figure 6: Cluster view of k-means node

Colour manager

Colourful and graphically creative clusters can be created using the colour manager node. Colours can be assigned for either nominal (possible values have to be available) or numeric columns (with lower and upper bounds).

Shape manager

This node assigns (different) shapes for each attribute value of one nominal column, i.e., for each possible value. It supports views, and then renders data points with the shape associated with the corresponding attribute value.

Using the shape manager node, three different shapes have been assigned to the clusters.

Cluster assigner

This node assigns new data to an existing set of prototypes, which are obtained by k-means clustering. Each data point is assigned to its nearest prototype.

Scatter plot

An interactive output view is created using a scatter plot node. It creates a scatter plot of two selectable attributes. Then each